Research Connect

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Intelligent Research Paper Search and Recommendation System

Data Structures and Algorithms

Abstract

Research Connect is an intelligent research paper search and recommendation system designed to address the growing challenge of accessing relevant academic literature from an ever-expanding collection of research publications. Traditional search methods often struggle to deliver accurate results when users provide incomplete, ambiguous, or misspelled queries. The proposed system leverages advanced searching and indexing techniques to enable efficient retrieval of research papers based on title, author, keywords, abstract content, and research domain. By integrating fuzzy search mechanisms and optimized pattern-matching algorithms, the platform significantly improves search accuracy and enhances the overall user experience.

Abstract

In addition to paper retrieval, Research Connect incorporates recommendation and analytical features that assist users in discovering related research work, identifying influential authors, and understanding citation relationships within the academic ecosystem. The system employs data structures and algorithms such as KMP, Rabin–Karp, Z Algorithm, Edit Distance, and graph-based analysis to provide intelligent search suggestions, citation analysis, and personalized recommendations. By combining efficient information retrieval with research analytics, Research Connect aims to create a centralized, scalable, and user-friendly platform that supports students, researchers, and academic institutions in conducting faster, smarter, and more effective research.

Problem Statement

01 Every year, thousands of research papers are published, making it difficult for students and researchers to find the right papers quickly.

02 Many search systems cannot handle spelling mistakes or incomplete keywords, which leads to poor search results.

03 Researchers often find it hard to discover related papers, important authors, and new research trends.

04 Research papers are spread across different sources, making it difficult to search, organize, and analyze information in one place.

Proposed Functionalities

Core Objective

To make research paper discovery faster and easier through intelligent search, recommendations, and research analytics.

- **Smart Paper Search**
  Search research papers by title, author, keywords, abstract, and domain.
- **Fuzzy Search**
  Find relevant papers even when users enter misspelled or incomplete keywords.
- **Paper Recommendations**
  Suggest related research papers based on user searches and paper similarity.
- **Citation Analysis**
  View citation counts and relationships between research papers and authors.
- **Research Analytics Dashboard**
  Analyze research trends, popular domains, and top authors through visual insights.

Tools and Technologies

01 **Programming Language**
Java for implementing the application, search algorithms, and backend logic.

02 **Database**
MySQL for storing research papers, author details, citations, and user information.

03 **Frontend & Backend**
HTML, CSS, JavaScript, and Spring Boot for building an interactive and user-friendly platform.

04 **DSA Algorithms**
Naive Search, KMP, Rabin-Karp, Z Algorithm, Edit Distance, and Graph Algorithms for efficient searching and recommendations.

Expected Outcomes

- Faster and more accurate research paper searching.
- Easy discovery of related papers and research topics.
- Better understanding of citations and author contributions.
- Improved user experience through smart search and recommendations.
- Practical application of Data Structures and Algorithms in a real-world system.

Key Impact

Research Connect is expected to significantly reduce literature review time and improve the quality of academic discovery through intelligent, algorithm-driven recommendations.